

Human-in-the-Loop (HITL) influence the AI lifecycle

Mapping Across Scenarios

S2 — Segmentation

AI Output

CT Study



Segmentation Model



Heart Structures

Human Action

Clinician:

- accepts segmentation
- corrects segmentation
- rejects segmentation

Example:

- **AI:** Left ventricle boundary = 92% correct
- **Doctor:** Adjusts boundary manually

The System capture:

- Original Segmentation
- Modified Segmentation
- Difference
- User
- Timestamp

S3 — 3D Reconstruction

AI Output

Segmented Heart



3D Model

Human Action

Clinician reviews:

- geometry
- completeness
- missing structures

Possible actions:

- Approve
- Request regeneration
- Flag issue

As learning Opportunity, this is less useful for direct model training.

More useful for:

- quality metrics
- pipeline optimization
- reconstruction parameter tuning

S4 — Clinical Decision Support

AI Output

Possible Coronary Stenosis

Confidence = 84%

Human Action

Possible feedback for Doctor:

- Confirmed
- False Positive
- Missed Finding

This becomes labeled clinical feedback.

Feedback Types

It's quite clear that not all feedback has equal value.

Feedback Type	Example	Learning Value
Accept	AI correct	Medium
Reject	AI wrong	High

Correction	Boundary adjusted	Very High
New Annotation	Missed lesion	Extremely High

Mapping Across Layer

Edge Layer

The clinician works here.

AI Result



Review UI



Validation

Everything occurs inside the hospital boundary.

Cloud Layer

Only receives *Validated Feedback Package*

Example:

- CaseID
- ModelVersion
- OriginalPrediction
- Correction
- ValidationOutcome

AI Maturity Roadmap

Phase	Action
Human validates AI output.	AI → Human
Feedback collected systematically.	AI → Human → Feedback Repository

Feedback enriches datasets.	AI → Human → Dataset Growth
Automated retraining pipeline.	Feedback ↓ Retraining ↓ Candidate Model
Governed model promotion.	Candidate Model ↓ Clinical Review Board ↓ Production

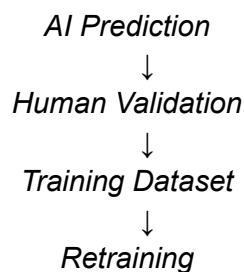
Maturity levels

Someone could argue that if the only way to update the model is to wait a human feedback, then the time for to put into operation an adequate model could be potentially high. In effect this is one of the fundamental trade-offs in healthcare AI.

But human-in-the-Loop does not necessarily mean that the model can improve only through explicit human corrections.

Level 1 — Pure Human-Supervised Learning

This is what we described so far.



Advantages:

- highest quality labels
- highest trust
- easiest to defend to regulators

Disadvantages:

- slow feedback accumulation

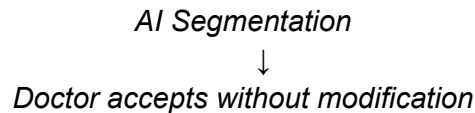
- expensive expert time
- long improvement cycles

This is typically where medical AI products start.

Level 2 — Human-Governed Learning

Now the system begins collecting much more information.

Example:



Even if the doctor does not draw a new contour, that acceptance itself becomes a signal.

Similarly:

1. Doctor opens result
2. Doctor spends 2 seconds
3. Doctor rejects

versus

1. Doctor spends 15 minutes
2. Doctor modifies extensively

These are weak signals but still valuable.

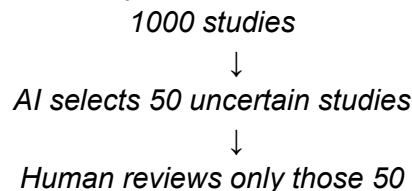
The platform can learn from:

- acceptance rates
- rejection rates
- correction frequency
- model confidence distributions

without waiting for perfect annotations every time.

Level 3 — Active Learning

The AI does not ask for feedback on every case. Instead:



The model is essentially saying:

"These are the cases from which I can learn the most."

This dramatically reduces expert effort.

In medical imaging, active learning is one of the most practical techniques to accelerate dataset growth.

Level 4 — Continuous Dataset Expansion

Suppose the platform processes 100,000 studies/year.

Even if only 2% receive detailed corrections, that's 2,000 expert-reviewed studies/year.

For many medical imaging applications, that is already an extremely valuable dataset.

The bottleneck in healthcare AI is often not model architecture, it is obtaining high-quality labeled data.

Possible Evolution

Most successful healthcare AI companies use a hybrid strategy:

Human Corrections
+
Human Approvals
+
Operational Metrics
+
Periodic Dataset Reviews

rather than relying only on explicit annotations.

The platform could prioritize clinically governed learning over autonomous self-optimization.

Model evolution would be driven by validated human expertise and controlled feedback loops, ensuring that improvements in predictive performance do not compromise traceability, safety, or regulatory compliance.

Feedback Intelligence Service

We could study the opportunity of a new service, between S5 and S6.

Validation
↓
Feedback Intelligence
↓
Learning Repository
↓
Training Dataset

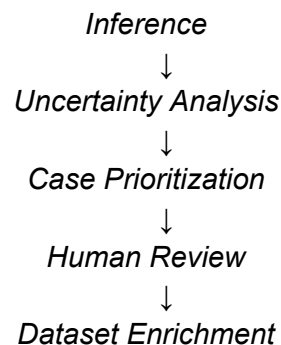
Its purpose is to determine:

- Which feedback is useful?
- Which feedback is noise?
- Which feedback should trigger retraining?

this service classifies feedback into:

Feedback	Learning Value
Accepted without changes	Low
Minor correction	Medium
Major correction	High
New annotation	Very High
AI rejection	Very High

Only the most valuable samples become retraining candidates.
After sufficient adoption, the system could introduce Active Learning.



At that point, clinicians are no longer reviewing cases merely to operate the platform.
They are reviewing the cases that maximize AI improvement.

The Real Cost

The cost is not primarily slower learning. The real cost is:

You cannot allow the AI to autonomously redefine what "correct" means.

In healthcare, the source of truth must remain a qualified clinician (or a formally validated dataset). That's why a mature healthcare AI architecture is usually closer to "AI assists humans" and "Humans improve AI", than to "AI improves itself"

The second model can converge faster, but it introduces risks such as model drift, silent degradation, and regulatory challenges that are often unacceptable in clinical environments.